

KRXGEN'26

GENERAL RULES AND REGULATIONS

INSTRUCTION:

- Per Team 3000 rupees.
- Team Size (3-4 Members).
- Per college maximum two teams will be allowed.
- Fill the google form, pay the Registration amount and kindly upload the screenshot in the given google form.
- For any doubts and clarification, kindly call to the staff and student coordinator and their numbers is mentioned on the Notice.
- Any other problem statements can also be taken apart from the given statements.

PROJECT DEVELOPMENT RULES:

- Participants must not present already existing projects.
- Every team must develop a new and innovative project within 24 hours during the hackathon.
- Participants should bring their own requirements and components needed for prototype development.
- Each team must develop a working prototype as the final output.
- The prototype can be developed in Hardware or Software.
- Projects should demonstrate innovation, creativity, technical feasibility, and real-world application.
- Teams must be prepared to explain and demonstrate their project during the final evaluation.
- Participants can create their own problem statement and create solution for the given theme.

THINGS TO BRING

Participants are requested to bring:

- Laptop
- Extension Box
- Pen Drive
- Required components and materials for prototype development
- Any other tools or equipment required for the project

DRESS CODE

- Participants are permitted to attend the hackathon in casual dress.
- Wear comfortable clothing suitable for 24-hour project development and prototype building.
- ID card is necessary.

ADDITIONAL GUIDELINES

- Come with your own innovative idea and be ready to turn it into a working prototype.
- Teams are encouraged to use readily available components, open-source technologies, and creative solutions.
- Ensure that your project is safe, functional, and demonstrable.
- Teamwork, innovation, and problem-solving will be key to success

BUILD. INNOVATE. CREATE.

24 HOURS • ONE CHALLENGE • ENDLESS POSSIBILITIES

THEME 1: SUSTAINABLE ENERGY TECHNOLOGY

KRXGEN'26 – RULES & REGULATIONS

THEME OBJECTIVE:

Develop innovative, practical, and sustainable engineering solutions that improve energy generation, storage, distribution, efficiency, and environmental sustainability.

ELIGIBLE PROBLEM AREAS

Participants may choose any one of the following:

- Renewable Energy Systems (Solar, Wind, Biomass, Hydro, Geothermal)
- Smart Energy Management
- Energy Storage Systems (Battery, Supercapacitor, Hydrogen)
- Electric Vehicle Charging Solutions
- Waste-to-Energy Technologies
- Green Manufacturing
- Carbon Footprint Reduction
- Smart Grid Technologies
- Energy-efficient Buildings
- AI/IoT for Energy Optimization
- Sustainable Agriculture Energy Solutions
- Rural Electrification
- Climate Change

RULES AND REGULATIONS

1. Team Formation

- Each team shall consist of 3–4 members.

- Interdisciplinary teams are encouraged.

2. Prototype Requirement

- Teams should develop a working prototype, simulation, software model, or proof-of-concept.
- Existing commercial products or copied projects are not permitted.

3. Innovation

- The solution must be original.
- Projects should address a real-world sustainability problem.
- Open-source libraries may be used with proper acknowledgment.

4. Technical Scope

Projects may include:

- Mechanical Systems
- Electrical Systems
- Electronics
- IoT
- AI/ML
- Robotics
- Embedded Systems
- Mobile/Web Applications
- CAD/CAE Simulation
- Digital Twin Technologies

5. Sustainability Criteria

Projects should demonstrate at least one of the following:

- Energy Saving
- Renewable Energy Utilization
- Pollution Reduction
- Carbon Emission Reduction

- Resource Conservation
- Circular Economy
- Sustainable Manufacturing

6. Deliverables

Each team must submit:

- Abstract
- Problem Statement
- Proposed Solution
- PPT Presentation
- Working Prototype/Simulation
- Source Code
- Bill of Materials (BOM)

7. Safety

- Unsafe electrical systems are prohibited.
- High-voltage equipment must have proper insulation.
- No hazardous chemicals or explosive materials are allowed.
- Teams are responsible for the safe operation of their prototypes.

8. Presentation

Each team will receive:

- 5 minutes for presentation.
- 5 minutes for Q&A with judges.

GENERAL GUIDELINES

- Bring your own laptop and required hardware.
- Internet access will be provided if available.
- Teams must carry valid college ID cards.

- Judges' decisions will be final and binding.

EXPECTED OUTCOMES

Participants should demonstrate:

- Engineering Innovation
- Practical Problem Solving
- Sustainable Development
- Technical Excellence
- Team Collaboration
- Social & Environmental Impact

FOR CLARIFICATION:

Mr. P. DHANUSH - 80728 35938

Mr. C. JEBI GODSON - 94438 50354

KRXGEN'26 – RULES & REGULATIONS

THEME 2: AUGMENTED REALITY & VIRTUAL REALITY (AR/VR) TECHNOLOGY

THEME OBJECTIVE:

Develop innovative, practical, and immersive AR/VR-based engineering solutions that enhance visualization, simulation, training, interaction, and problem-solving across various domains.

PROBLEM AREAS

- Open Theme

RULES AND REGULATIONS

1. Team Formation

- Teams must develop a working AR/VR prototype, simulation, or immersive application.
- Existing commercial products or copied projects are not permitted.
- Projects must be demonstrable during the evaluation.

2. Prototype Requirement

- The solution must be original and innovative.
- Projects should address a real-world problem using AR, VR, or XR technology.
- Open-source assets, libraries, or SDKs may be used with proper acknowledgment.

3. Technical Scope

Projects may include one or more of the following:

- Augmented Reality (AR)
- Virtual Reality (VR)
- Mixed Reality (XR)
- Unity Game Engine
- OpenXR
- AR Foundation
- ARCore / ARKit
- AI Integration
- Mobile, PC, or VR-based applications

4. Project Evaluation Focus

Projects should demonstrate one or more of the following:

- Real-world impact
- Innovation and creativity
- Practical usability
- Immersive user experience
- Technical implementation
- Scalability and future scope

5. Deliverables

Each team must submit:

- Project Abstract
- Problem Statement
- Proposed Solution
- PPT Presentation
- Working AR/VR Prototype
- Source Code (if requested)
- Demo Video or Project Link (optional)

6. Safety & Ethics

- Projects must not contain harmful, offensive, or inappropriate content.
- Applications should not misuse personal or sensitive data.
- Participants are responsible for the safe operation of their devices during demonstrations.
- Teams must assist users while demonstrating VR experiences to ensure participant safety.

7. Presentation

Each team will receive:

- 5 minutes for project demonstration.
- 5 minutes for technical explanation and Q&A.

GENERAL GUIDELINES

- Bring your own laptop and required hardware.
- AR demonstrations must use participants' own smartphones.
- Teams must carry valid college ID cards.
- Projects must run successfully on the demonstration device.
- Judges' decisions shall be final and binding.

**VR DEVICES WILL NOT BE PROVIDED TO PARTICIPANTS. THE
FINAL OUTCOMES WILL BE EVALUATED BY THE JUDGES
USING VR DEVICES.**

EXPECTED OUTCOMES

Participants should demonstrate:

- Innovation in AR/VR/XR
- Practical problem solving
- Effective use of immersive technologies
- Technical excellence
- Creativity and user experience
- Team collaboration
- Real-world impact

**VR DEVICES WILL NOT BE PROVIDED TO PARTICIPANTS. THE
FINAL OUTCOMES WILL BE EVALUATED BY THE JUDGES
USING VR DEVICES.**

FOR CLARIFICATION:

Mr. A. GOWTHAM - 8608584526
Mr. S. DHAMODHARAN DOSS - 9080964118

KRXGEN'26 – RULES & REGULATIONS

THEME 3: INDUSTRY 5.0 AND 6.0

THEME OBJECTIVE:

Industry 5.0 and Industry 6.0: Transforming Manufacturing through AI, Sustainability, and Autonomous Intelligence

ELIGIBLE PROBLEM AREAS

Participants may choose any one of the following:

- Human–Robot Collaboration (Cobots)
- Cybersecurity in Smart Factories
- Artificial Intelligence in Manufacturing
- Sustainable and Green Manufacturing
- Digital Twin Technology
- Industrial Internet of Things (IIoT)
- Autonomous Manufacturing Systems (Industry 6.0)
- Supply Chain Resilience
- Energy Management
- Machine Vision for Defect Detection
- Quantum Computing for Industrial Optimization

RULES AND REGULATIONS

1. Team Formation

- Each team shall consist of 3–4 members.
- Students from any AICTE-approved institution are eligible.
- Interdisciplinary teams are encouraged.

2. Prototype Requirement

- Teams should develop a working prototype, simulation, software model, or proof-of-concept.
- Existing commercial products or copied projects are not permitted.

3. Innovation

- The solution must be original.

- Projects should address a real-world sustainability problem.
- Open-source libraries may be used with proper acknowledgment.

4. Technical Scope

Projects may include:

- Human–Robot Collaboration (Cobots)
- Artificial Intelligence (AI) and Machine Learning (ML)
- Industrial Internet of Things (IIoT)
- Digital Twin Technology
- Cyber-Physical Systems (CPS)
- Smart Manufacturing and Industry Automation
- Big Data Analytics
- Cloud and Edge Computing
- Additive Manufacturing (3D Printing)
- Augmented Reality (AR) and Virtual Reality (VR)
- Predictive Maintenance
- Intelligent Quality Control using Computer Vision
- Sustainable and Green Manufacturing

5. Sustainability Criteria

Projects should demonstrate at least one of the following:

- Energy Saving
- Renewable Energy Utilization
- Pollution Reduction
- Carbon Emission Reduction
- Resource Conservation
- Circular Economy
- Sustainable Manufacturing

6. Deliverables

Each team must submit:

- Abstract (Maximum 300 words)

- Problem Statement
- Proposed Solution
 - PPT Presentation
 - Working Prototype/Simulation
- Source Code
- Bill of Materials (BOM)

7. Safety

- Unsafe electrical systems are prohibited.
- High-voltage equipment must have proper insulation.
- No hazardous chemicals or explosive materials are allowed.
- Teams are responsible for the safe operation of their prototypes.

8. Presentation

Each team will receive:

- 5 minutes for presentation.
- 5 minutes for Q&A with judges.

GENERAL GUIDELINES

- Bring your own laptop and required hardware.
- Internet access will be provided if available.
- Teams must carry valid college ID cards.
- Any form of plagiarism will lead to immediate disqualification.
- Judges' decisions will be final and binding.
- Organizers reserve the right to modify the schedule if necessary.

EXPECTED OUTCOMES

Participants should demonstrate:

- Engineering Innovation
- Practical Problem Solving
- Sustainable Development
- Technical Excellence
- Team Collaboration
- Social & Environmental Impact

FOR CLARIFICATION:

Mr. E. LARSHAD -9080926625

Mr. B. MATHAN -6374751

